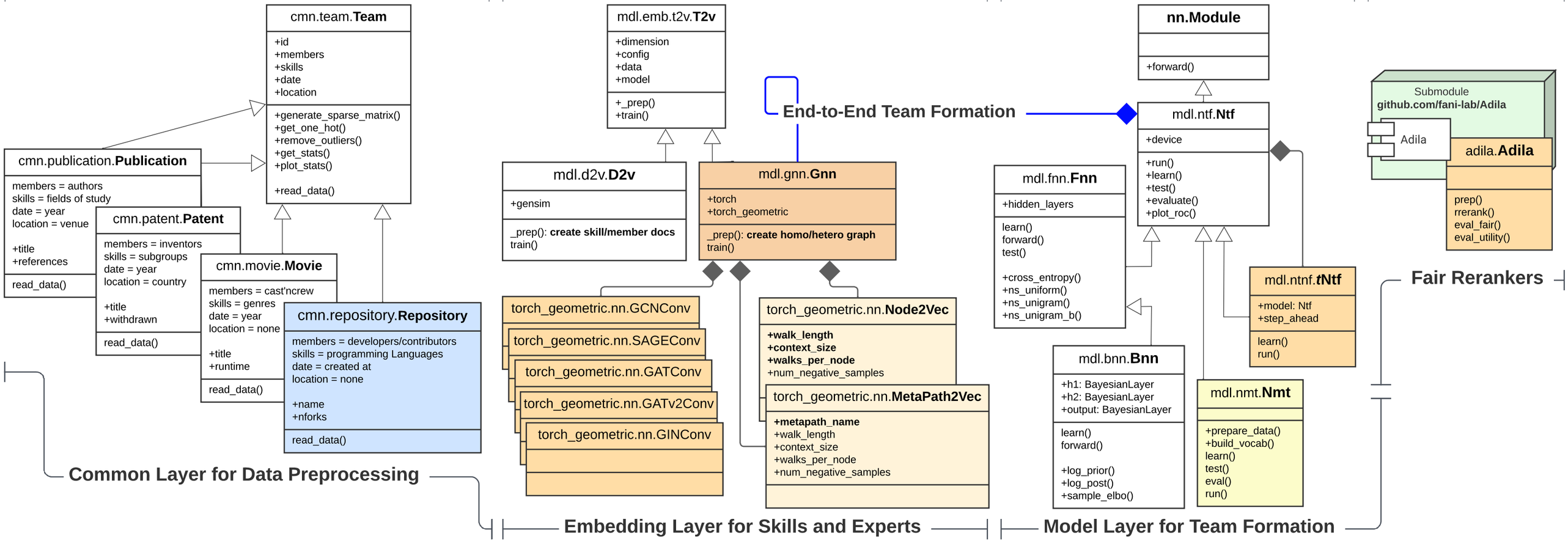

cmn.team.Team
+id
+members
+skills
+date
+location
+generate_sparse_matrix()
+get_one_hot()
+remove_outliers()
+get_stats()
+plot_stats()
+read_data()
cmn.publication.Publication
members = authors
skills = fields of study
date = year
location = venue
+title
+references
read_data()
cmn.patent.Patent
members = inventors
skills = subgroups
date = year
location = country
+title
+withdrawn
read_data()
cmn.movie.Movie
members = cast'ncrew
skills = genres
date = year
location = none
+title
+runtime
read_data()
cmn.repository.Repository
members = developers/contributors
skills = programming Languages
date = created at
location = none
+name
+nforks
read_data()
Common Layer for Data Preprocessing
mdl.emb.t2v.T2v
+dimension
+config
+data
+model
+_prep()
+train()
End-to-End Team Formation
mdl.d2v.D2v
+gensim
_prep(): create skill/member docs
train()
mdl.gnn.Gnn
+torch
+torch_geometric
_prep(): create homo/hetero graph
train()
torch_geometric.nn.GCNConv
torch_geometric.nn.SAGEConv
torch_geometric.nn.GATConv
torch_geometric.nn.GATv2Conv
torch_geometric.nn.GINConv
torch_geometric.nn.Node2Vec
+walk_length
+context_size
+walks_per_node
+num_negative_samples
torch_geometric.nn.MetaPath2Vec
+metapath_name
+walk_length
+context_size
+walks_per_node
+num_negative_samples
Embedding Layer for Skills and Experts
nn.Module
+forward()
mdl.ntf.Ntf
+device
+run()
+learn()
+test()
+evaluate()
+plot_roc()
mdl.fnn.Fnn
+hidden_layers
learn()
forward()
test()
+cross_entropy()
+ns_uniform()
+ns_unigram()
+ns_unigram_b()
mdl.ntnf.tNtf
+model: Ntf
+step_ahead
learn()
run()
mdl.bnn.Bnn
+h1: BayesianLayer
+h2: BayesianLayer
+output: BayesianLayer
learn()
forward()
+log_prior()
+log_post()
+sample_elbo()
mdl.nmt.Nmt
+prepare_data()
+build_vocab()
learn()
test()
eval()
run()
Model Layer for Team Formation
Submodule
github.com/fani-lab/Adila
Adila
adila.Adila
prep()
rrerank()
eval_fair()
eval_utility()
Fair Rerankers